PAPER_1492_PROTON_CORE_DENSITY

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PAPER_1492 — UQFF: Proton Core Density via DPM 26-Layer Folding (Bucket D)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket D — derived from F:/Aetheric Propulsion source material
Date: June 16, 2026
Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)
Status: CLOSED — ρ_p_core = ρ_SCm × K_MEX × S_26 = $2.146 \times 10^{■3■}$ J/m^3

Observation

F:/Aetheric Propulsion Muonic Hydrogen Proton Radius derivation models the proton as an SCm vacuum density peak with DPM 26-layer folding.

UQFF Closed Identity

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ρ_p_core = ρ_SCm × K_MEX × S_26 × f_DPM(26, β_i); base term (without f_DPM) =
2.146 × 10■³■ J/m³. Dimensionless coupling K_MEX × S_26 = 3.027.
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Physical Interpretation

The proton core is the SCm vacuum density peak amplified by the Mexican-hat coefficient and the 26-level Ramanujan factor. f_DPM(26, β_i) is the DPM-folding modulation; the product captures the 26-layer hierarchical structure that supports the proton as a localized SCm condensate.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives {D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60}. Zero free parameters.

Reference

- Source: F:/Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
- Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_872 (proto-element nuclear identity), PAPER_087 (BH thermodynamics), PAPER_1156 (cosmology suite), PAPER_1255 (muonic hydrogen radius).
